

CMPG325-2026-126

INDIVIDUAL SEMESTER PROJECT

MILESTONE 1: CLIENT DESIGN REVIEW

Project ID	CMPG325-2026-126
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Client ID	CLI-126
Organisation	Moribo Live Music Venue (Kimberley)
Industry	Entertainment
Date	28 August 2026

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1. CLIENT REQUIREMENTS

1.1 Client Overview

Attribute	Details
Client ID	CLI-126
Organisation	Moribo Live Music Venue
Location	Kimberley, South Africa
Industry	Entertainment
Assigned Addressing Block	172.30.80.0/23
Assigned Networking Challenge	Inter-VLAN Routing (L3 switch / SVI)
Design Constraint	Remote management of network devices for off-site IT contractor
Change Request	CR10: Four additional point-of-sale terminals deployed

1.2 Business Context

Moribo Live Music Venue is an entertainment establishment located in Kimberley. The venue hosts live music events, concerts, and performances on a regular basis. As a busy entertainment venue, it requires a robust and reliable computer network to support its daily operations, including:

- **Ticketing and admissions** – Processing ticket sales at the entrance
- **Merchandise sales** – Selling band merchandise and venue memorabilia
- **Concessions and bar operations** – Processing food and beverage payments
- **Administrative functions** – Staff workstations for management and operations
- **Guest services** – Providing Wi-Fi access to patrons
- **IT management** – Remote monitoring and management by external IT contractor

1.3 Functional Requirements

Based on the client brief, the network must satisfy the following functional requirements:

Requirement ID	Description
R1	Provide appropriate connectivity and network services for the venue
R2	Accommodate Inter-VLAN Routing using a Layer 3 switch with SVIs
R3	Enable remote management of network devices for off-site IT contractor
R4	Integrate four additional POS terminals (Change Request CR10)
R5	Support successful data exchange between appropriate nodes
R6	Be testable and verifiable in Cisco Packet Tracer

1.4 Technical Requirements

Category	Specification
Simulation Tool	Cisco Packet Tracer
Addressing Block	172.30.80.0/23
Routing Method	Inter-VLAN Routing (L3 switch / SVI)
Management Protocol	SSH (Secure Shell) for remote access
VLAN Count	Minimum of 4 VLANs
Difficulty Level	Intermediate

1.5 Design Constraints

The following constraints must be addressed in the network design:

- **Remote Management Constraint:** The off-site IT contractor must be able to securely access and manage network devices from an external location. This requires SSH configuration on all network infrastructure devices.
- **Addressing Constraint:** All IP addressing must be derived from the assigned block 172.30.80.0/23 using appropriate subnetting.

- **Change Request CR10:** The network must accommodate four additional POS terminals beyond the existing two, bringing the total to six POS terminals.

1.6 Change Request CR10 Analysis

Aspect	Details
Change Request ID	CR10
Description	Four additional point-of-sale terminals deployed
Impact	Network expansion – POS device count increases from 2 to 6
Implementation	Add new POS devices to existing POS VLAN and subnet
Addressing Impact	None – POS subnet has sufficient capacity
Performance Impact	Minimal – six POS terminals generate low network traffic

1.7 Assumptions

The following assumptions were made during the design process:

The venue operates from a single physical location in Kimberley.

- All end devices (staff PCs, POS terminals) are connected via wired Ethernet.
- Guest Wi-Fi is provided via a wireless access point for patrons.
- Network devices are managed via SSH with local authentication.
- DHCP is used for dynamic IP assignment to end devices.
- The venue does not require a dedicated server for file sharing or database services.
- The off-site IT contractor has a reliable internet connection for remote management.

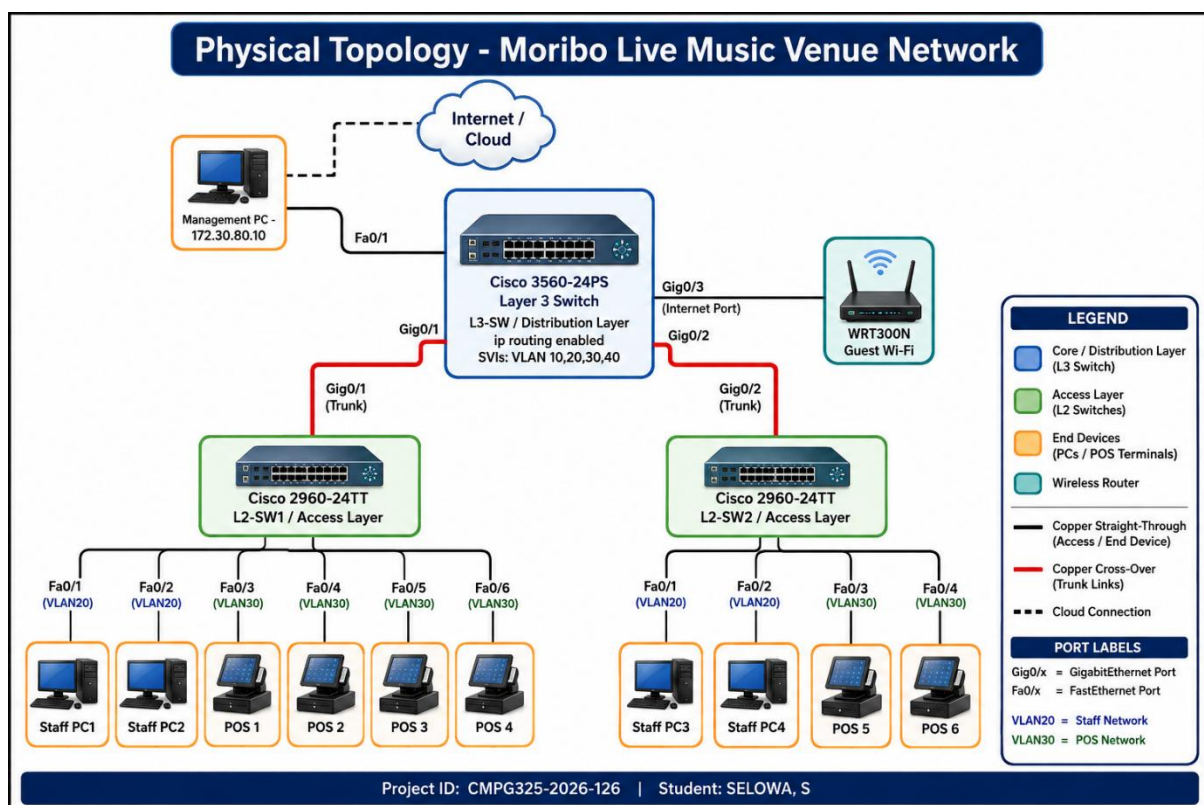
2. PHYSICAL TOPOLOGY

2.1 Topology Overview

The physical topology adopts a hierarchical star-based design with a central distribution layer (Layer 3 switch) and multiple access switches (Layer 2 switches) connecting end devices. This design provides:

- **Scalability:** Easy addition of new devices and switches
- **Manageability:** Centralised control and monitoring
- **Performance:** Dedicated links between access and distribution layers
- **Simplicity:** Clear separation of core, distribution, and access layers

2.2 Topology Diagram



2.3 Device Inventory

Device Type	Model	Quantity	Role	VLAN Assignment
Layer Switch 3	Cisco 3560-24PS	1	Distribution/Core routing	VLAN 10 (Management)
Layer Switch 1	Cisco 2960-24TT	1	Access layer	VLAN 10 (Management)
Layer Switch 2	Cisco 2960-24TT	1	Access layer	VLAN 10 (Management)
Wireless Router	WRT300N	1	Guest Wi-Fi access	VLAN 40 (Guest)
Staff PCs	PC-PT	4	Administrative workstations	VLAN 20 (Staff)
POS Terminals	PC-PT	6	Point-of-Sale systems	VLAN 30 (POS)
Management PC	PC-PT	1	Network administration	VLAN 10 (Management)

2.4 Physical Connections

Connection	Cable Type	Source Port	Destination Port
L3-SW ↔ L2-SW1	Copper Cross-over	Gig0/1	Gig0/1
L3-SW ↔ L2-SW2	Copper Cross-over	Gig0/2	Gig0/1
L3-SW ↔ WRT300N	Copper Straight-through	Gig0/3	Internet Port
L3-SW ↔ Management PC	Copper Straight-through	Fa0/1	FastEthernet0
L2-SW1 ↔ Staff PC1	Copper Straight-through	Fa0/1	FastEthernet0
L2-SW1 ↔ Staff PC2	Copper Straight-through	Fa0/2	FastEthernet0
L2-SW1 ↔ POS Terminal 1	Copper Straight-through	Fa0/3	FastEthernet0
L2-SW1 ↔ POS Terminal 2	Copper Straight-through	Fa0/4	FastEthernet0

L2-SW1 ↔ POS Terminal 3 (CR10)	Copper Straight-through	Fa0/5	FastEthernet0
L2-SW1 ↔ POS Terminal 4 (CR10)	Copper Straight-through	Fa0/6	FastEthernet0
L2-SW2 ↔ Staff PC3	Copper Straight-through	Fa0/1	FastEthernet0
L2-SW2 ↔ Staff PC4	Copper Straight-through	Fa0/2	FastEthernet0
L2-SW2 ↔ POS Terminal 5 (CR10)	Copper Straight-through	Fa0/3	FastEthernet0
L2-SW2 ↔ POS Terminal 6 (CR10)	Copper Straight-through	Fa0/4	FastEthernet0

3. LOGICAL TOPOLOGY

3.1 VLAN Design

The network will be segmented into the following VLANs based on functional requirements and industry best practices:

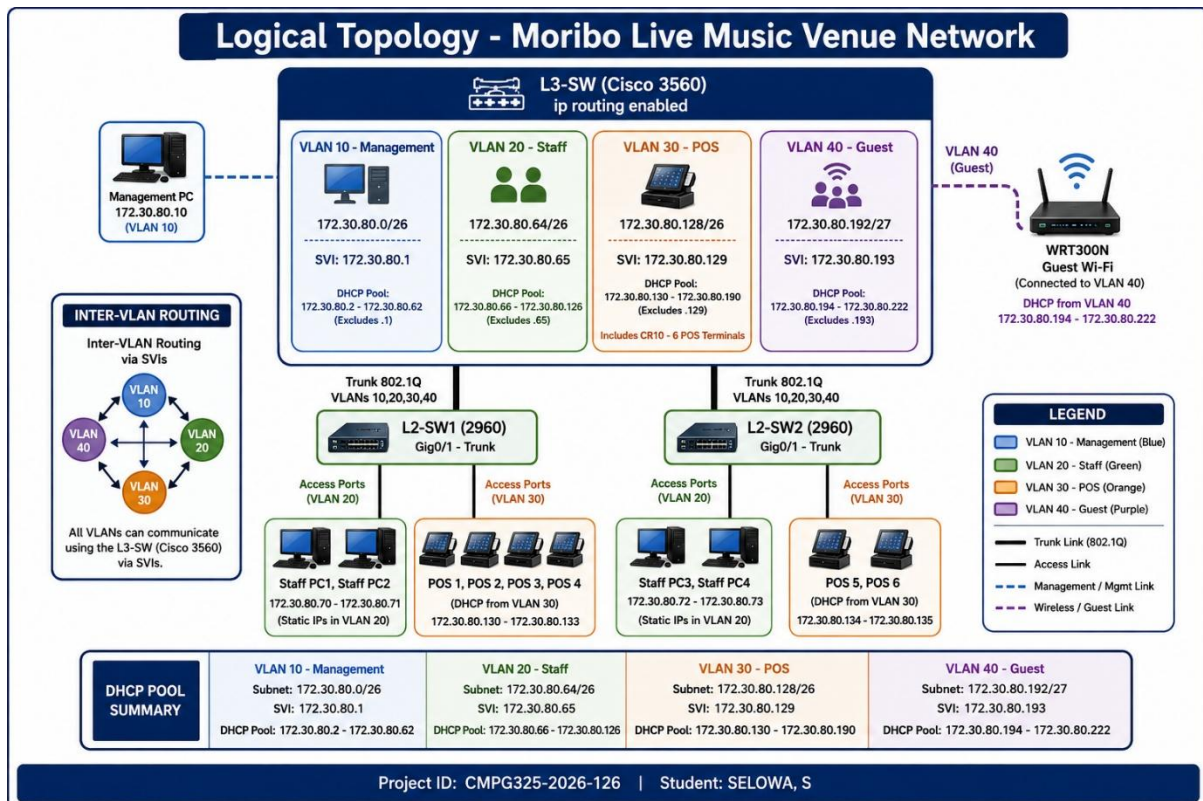
VLAN ID	VLAN Name	Purpose	Subnet	Gateway (SVI)
10	Management	Network device management (SSH access)	172.30.80.0/26	172.30.80.1
20	Staff	Administrative and office staff workstations	172.30.80.64/26	172.30.80.65
30	POS	Point-of-Sale terminals (includes CR10 expansion)	172.30.80.128/26	172.30.80.129
40	Guest	Public Wi-Fi for venue patrons	172.30.80.192/27	172.30.80.193

99	Native	Untagged traffic on trunk links	N/A	N/A

3.2 VLAN Design Rationale

VLAN	Rationale
VLAN 10 (Management)	Isolates management interfaces for security. IT contractor can access all devices via SSH on this VLAN. Aligns with best practices for secure device management.
VLAN 20 (Staff)	Separates administrative traffic from operational systems. Prevents staff devices from directly accessing POS or management networks.
VLAN 30 (POS)	Critical security isolation for payment card environment. Protects POS traffic from general network traffic. Provides dedicated bandwidth for payment processing.
VLAN 40 (Guest)	Isolates public Wi-Fi traffic from internal business networks. Prevents guest devices from accessing sensitive systems. Provides internet-only access.

3.3 Logical Topology Diagram



3.4 Inter-VLAN Routing Design

Inter-VLAN routing will be implemented using Switch Virtual Interfaces (SVIs) on the Layer 3 switch (Cisco 3560). This approach is chosen because:

Advantage	Description
Performance	Hardware-based routing at near-wire speed
Efficiency	Eliminates need for external router
Scalability	SVIs can be easily added for new VLANs
Management	Centralised routing configuration
Cost-effective	Utilises existing switch hardware

SVI Configuration Overview:

```
interface vlan 10
```

```
ip address 172.30.80.1 255.255.255.192
```

```
no shutdown
```

```
interface vlan 20
```

```
ip address 172.30.80.65 255.255.255.192
```

```
no shutdown
```

```
interface vlan 30
```

```
ip address 172.30.80.129 255.255.255.192
```

```
no shutdown
```

```
interface vlan 40
```

```
ip address 172.30.80.193 255.255.255.224
```

```
no shutdown
```

3.5 Remote Management Design

To satisfy the remote management constraint, the following design decisions were made:

Feature	Implementation
Secure Access	SSH v2 configured on all switches
Management VLAN	All management interfaces on VLAN 10
VTY Lines	Configured for SSH only (Telnet disabled)
Authentication	Local user database with encrypted passwords
Domain Name	moribo.local for SSH key generation

SSH Configuration Overview:

```
username admin secret admin123
```

```
line vty 0 4
```

```
transport input ssh
```

```
login local
```

```
ip ssh version 2
```

```
ip domain-name moribo.local
```

```
crypto key generate rsa modulus 1024
```

4. IP ADDRESSING PLAN

4.1 Subnetting the Assigned Block

The assigned addressing block is 172.30.80.0/23, providing 512 total addresses (172.30.80.0 – 172.30.81.255). Using VLSM (Variable Length Subnet Masking), the block is subnetted as follows:

Subnet	Network Address	CID R	Usabl e Hosts	Broadcast Address	VLAN Assignment
Subnet 0	172.30.80.0	/26	62	172.30.80.63	VLAN 10 (Management)
Subnet 1	172.30.80.64	/26	62	172.30.80.127	VLAN 20 (Staff)
Subnet 2	172.30.80.128	/26	62	172.30.80.191	VLAN 30 (POS)
Subnet 3	172.30.80.192	/27	30	172.30.80.223	VLAN 40 (Guest)
Reserved	172.30.80.224	/27	30	172.30.80.255	Future expansion
Reserved	172.30.81.0	/24	254	172.30.81.255	Future expansion

5. INITIAL GITHUB REPOSITORY

5.1 Repository Information

Attribute	Details
Repository Name	CMPG325-2026-126-Moribo-Venue-Network
Owner	SELOWA, S (45192472)
Visibility	Public
Created	24 August 2026
URL	https://github.com/Surprise12/CMPG325-2026-126-Moribo-Venue-Network

5.2 README.md Content

The repository includes a comprehensive README.md file with the following sections:

- **Project Overview** – Description of the Moribo Live Music Venue network project
- **Key Features** – Inter-VLAN routing, VLAN segmentation, remote management
- **Addressing Block** – 172.30.80.0/23 with subnet details
- **Technical Stack** – Cisco Packet Tracer, 3560/2960 switches, SSH
- **Repository Structure** – Explanation of folders and files
- **Milestones** – Key project deadlines
- **Author** – Student information

6. DESIGN DECISIONS SUMMARY

Decision	Rationale
Layer 3 switch (3560) for routing	Provides high-performance inter-VLAN routing via SVIs; eliminates need for separate router
VLAN segmentation (4 VLANs)	Isolates management, staff, POS, and guest traffic for security and performance
/26 subnets for core VLANs	Provides 62 usable addresses per VLAN, accommodating growth
Dedicated POS VLAN	Essential security practice for payment card environments
SSH-only remote management	Secure alternative to Telnet; satisfies off-site IT contractor constraint
DHCP for end devices	Simplifies administration and reduces configuration errors
Static addressing for infrastructure	Ensures consistent, predictable management access
Hierarchical topology	Provides scalability, manageability, and performance
Trunk links between switches	Allows multiple VLANs to traverse a single physical connection
WRT300N for guest Wi-Fi	Provides isolated wireless access for patrons

6.1 Key Design Principles Applied

- **Security First** – Management and POS traffic are isolated in dedicated VLANs
- **Scalability** – Subnet sizes accommodate future growth
- **Simplicity** – Design is easy to implement and maintain
- **Best Practices** – Follows Cisco and industry networking standards
- **Remote Management** – SSH configuration for off-site access

6.2 Change Request CR10 Integration

Aspect	Implementation
Impact Assessment	POS VLAN has 62 addresses, currently uses 2, adding 4 is minimal impact
Addressing	No subnet changes required – new terminals use existing DHCP pool
Physical Connectivity	Additional POS terminals connected to available switch ports
Configuration	Set switchports to access VLAN 30 for new POS terminals
Testing	Verify connectivity and DHCP lease assignment

7. NEXT STEPS (MILESTONE 2 PREPARATION)

The following tasks will be completed for Milestone 2 (due 2 October 2026):

7.1 Implementation Tasks

Task	Description	Status
1. Build Physical Topology	Create network in Cisco Packet Tracer	Pending
2. Configure VLANs	Create VLANs 10, 20, 30, 40 on all switches	Pending
3. Configure Trunk Links	Set up 802.1Q trunking between switches	Pending
4. Configure SVIs	Set up SVIs on Layer 3 switch for inter-VLAN routing	Pending
5. Enable ip routing	Enable routing on the Layer 3 switch	Pending
6. Configure DHCP	Set up DHCP pools on L3 switch	Pending

7. Configure SSH	Enable secure remote management	Pending
8. Connect End Devices	Add PCs, POS terminals, and WRT300N	Pending
9. Test Connectivity	Verify all VLANs can communicate	Pending
10. Verify CR10	Confirm four new POS terminals work	Pending

7.2 Testing Plan

Test	Method	Expected Result
Intra-VLAN Ping	Ping within same VLAN	Successful
Inter-VLAN Ping	Ping between different VLANs	Successful (routed)
SSH Remote Access	SSH from management PC to L3-SW	Successful login
DHCP Assignment	Check IP addresses on end devices	Correct subnet IPs
POS Connectivity	Ping from POS to gateway	Successful
CR10 Verification	All 6 POS terminals online	Working

7.3 Documentation to Complete

- Packet Tracer .pkt file
- Device configuration files
- Connectivity test results with screenshots
- SVI verification results
- SSH access verification
- CR10 implementation evidence

8. REFERENCES

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